

EXPERIMENT -10

Problem Statement

In Linux, the shell acts as an interface between the user and the operating system kernel. Different types of shells provide varied features, scripting capabilities, and command-line behaviors. Among these, Bash (Bourne Again Shell) is the most widely used shell due to its scripting power and ease of use.

The problem is to identify different types of shells available in Linux, understand their characteristics, and write and execute a simple shell script using Bash, including the use of variables

AIM

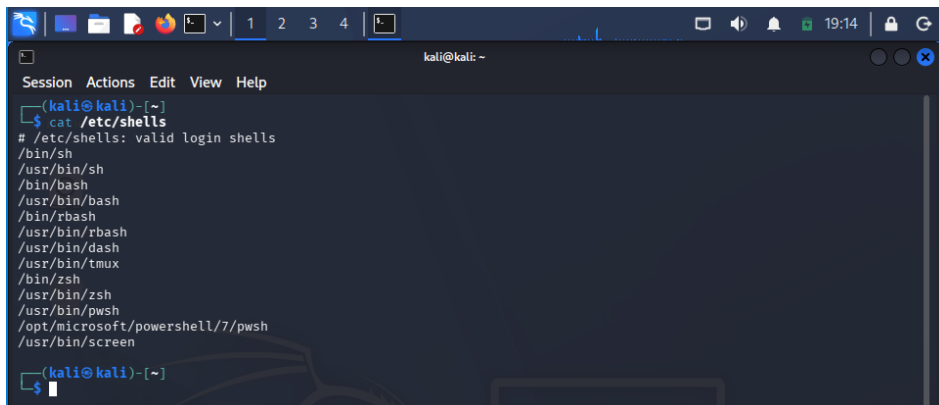
Understanding Different Types of Shell and Writing a Simple Shell Script.

PROCEDURE

A. Identifying Different Shells

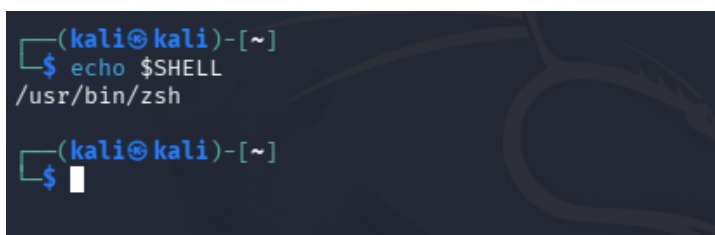
Open the terminal

List available shells: `cat /etc/shells`



```
(kali@kali)-[~]
$ cat /etc/shells
# /etc/shells: valid login shells
/bin/sh
/usr/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/usr/bin/dash
/usr/bin/tmux
/bin/zsh
/usr/bin/zsh
/usr/bin/pwsh
/opt/microsoft/powershell/7/pwsh
/usr/bin/screen
```

Check the current shell: `echo $SHELL`



```
(kali@kali)-[~]
$ echo $SHELL
/usr/bin/zsh
```

B. Switching Between Shells

Switch to Bourne shell: sh

Return to Bash: exit

```
(kali@kali)-[~]
$ echo $SHELL
/usr/bin/zsh

(kali@kali)-[~]
$ sh
$ exit

(kali@kali)-[~]
$
```

C. Writing A Simple Bash Script

Create a new script file: nano simple.sh

```
(kali@kali)-[~]
$ nano simple.sh

(kali@kali)-[~]
$
```

Enter the following script: `#!/bin/bash echo "Welcome to Linux Shell Scripting"`
`name="Student" echo "Hello $name"`

```
kali@kali: ~
Session Actions Edit View Help
GNU nano 8.7.1 simple.sh
#!/bin/bash
echo "Welcome to Linux Shell Scripting"
name="Student"
echo "Hello $name"
```

Save and exit the editor.

D. Executing the Script

Provide execute permission: `chmod +x simple.sh`

```
(kali㉿kali)-[~]  
$ chmod +x simple.sh  
  
(kali㉿kali)-[~]  
$
```

Run the script: ./simple.sh

```
(kali㉿kali)-[~]  
$ nano simple.sh  
  
(kali㉿kali)-[~]  
$ chmod +x simple.sh  
  
(kali㉿kali)-[~]  
$ ./simple.sh  
Welcome to Linux Shell Scripting  
Hello Student  
  
(kali㉿kali)-[~]  
$
```

Expected Output

- List of available shells displayed.
- Current shell identified.
- Simple Bash script executed successfully.
- Output displays variable values correctly.

Result

Different types of Linux shells were identified and a simple Bash shell script using variables was successfully written and executed.